

Kestrel downstream oxygen sensor aging triggering catalyst efficiency code

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Area: bulletins

1 Condition

Affected Kestrel vehicles may set DTC P0420 (Catalyst System Efficiency Below Threshold, Bank 1) with the MIL illuminated and no driveability complaint. The code typically appears after extended mileage as the post-catalyst signal begins to track the pre-catalyst signal too closely. Freeze-frame data usually shows steady-state cruise conditions at operating temperature when the fault matures. Background on this code is provided in Kestrel Emissions Service Manual :: Catalyst Efficiency Faults .

2 Cause

The most common cause is aging of the downstream oxygen sensor, part number O2D-5120, whose slowed switching response is misinterpreted by the monitor as reduced catalyst storage. In many cases the catalytic converter CAT-5500 is still functional, and a lazy O2D-5120 alone trips P0420. Sensor function and placement are described in Kestrel Emissions Service Manual :: Upstream and Downstream Oxygen Sensors , and converter theory in Kestrel Emissions Service Manual :: Catalytic Converter (Bank 1 and Bank 2) .

3 Repair Procedure

Replace the downstream oxygen sensor O2D-5120 following PROC-O2-SENSOR, applying anti-seize sparingly to the threads and torquing to specification. After replacement, clear DTC P0420 and run the catalyst monitor to completion to confirm the code does not return before condemning the converter. The detailed steps are in Kestrel Emissions Service Manual :: Oxygen Sensor Replacement Procedure , within the broader Kestrel Emissions Service Manual :: Oxygen Sensing section.

4 Catalyst Replacement (If Required)

If P0420 returns after the O2D-5120 sensor is replaced and verified, the catalytic converter CAT-5500 has degraded and must be replaced following PROC-CAT. Inspect for upstream causes such as misfire or oil consumption before fitting a new CAT-5500 to avoid repeat failure. The replacement procedure is documented in Kestrel Emissions Service Manual :: Catalytic Converter Replacement Procedure .